

I build reusable algorithmic components: AdvancedAlgorithms, PS-form memory-dependence analysis, and graph infrastructure for an LLVM-track internship.

**12 C++23 modules**  
graphs, trees, strings, and data structures

**1st · 104/104**  
PS-form; only maximum result among 49

**500k vertices**  
stress tests for recursion and accidental  $O(n^2)$

## EXPERIENCE

|                |                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 — PRESENT | <b>AdvancedAlgorithms</b> <ul style="list-style-type: none"><li>Explicit contracts, complexity notes, differential oracles, and structural invariants.</li><li>ASan/UBSan and separate correctness/performance suites.</li></ul> |
| JUL–AUG 2026   | <b>MCST</b> <ul style="list-style-type: none"><li>Shared graph model plus RPO/cycles, Dijkstra, Tarjan SCC, and Dinic.</li></ul>                                                                                                 |

## SELECTED PROJECTS

|                                                                                                        |
|--------------------------------------------------------------------------------------------------------|
| <b>PS-form Analyzer</b><br>Conservative memory-dependence analysis with exact and metamorphic testing. |
| <b>Wist SSA</b><br>Callable-first SSA, CFG/dominance verification, and descriptor-driven folding.      |
| <b>x86 Codegen Lab</b><br>Liveness, live intervals, and linear-scan register allocation.               |

## SKILLS

|                                                                                 |
|---------------------------------------------------------------------------------|
| <b>Algorithms</b><br>graphs, trees, strings, data structures, optimization      |
| <b>Correctness</b><br>differential tests, exact oracles, invariants, sanitizers |
| <b>Complexity</b><br>explicit bounds, large inputs, non-flaky performance gates |
| <b>Languages</b><br>C++23, C17, Python, C#/.NET, Rust                           |

## EDUCATION

Software Engineering student at HSE University

## RECOGNITION

Ranked 1st of 49 in MCST selection; HSE Vysshaya Proba prize-winner; top MEPhI Junior results in 2025/2026.

## TECHNICAL COMMUNICATION

Teaching algorithms and reviewing solutions since 2021.

## AVAILABILITY

Moscow / remote · up to 20 hours/week from September 2026